Skills Summary

Triangle Similarity — Reversed Correspondence and Scale-Factor Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading the correspondence from the statement

Rule: the letters of a similarity statement are matched *in order*. $\triangle ABC \sim \triangle DEF$ means $A \leftrightarrow D,\ \ B \leftrightarrow E,\ \ C \leftrightarrow F$, and therefore

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$

- A side is named by its two endpoints, so its partner is named by the two matching letters — $\overline{BC}$ pairs with $\overline{EF}$, never with $\overline{DE}$.
- Write the pairs down before writing any fraction. A scrambled statement such as $\triangle ABC \sim \triangle FED$ is legal and means $A \leftrightarrow F$, $B \leftrightarrow E$, $C \leftrightarrow D$.
- Keep every ratio pointing the same way: first triangle on top in *both* fractions, or second on top in both.

Example: $\triangle ABC \sim \triangle DEF$ with $AB = 4$, $BC = 7$ and $DE = 10$. Find EF.
Step 1. $\overline{AB}$ and $\overline{DE}$ are a corresponding pair and both lengths are known, so they fix the factor; $\overline{BC}$ pairs with the unknown $\overline{EF}$.
Step 2. DEF is the larger triangle $(10 > 4)$, so multiply by $\frac{10}{4}$: $EF = 7 \cdot \frac{10}{4}$. $EF = \mathbf{17.5}$, longer than BC as expected.

Try it: $\triangle ABC \sim \triangle DEF$ with $AB = 3$, $BC = 5$ and $DE = 12$. Find EF.

check: **20**

Skill 2 — Pointing the scale factor the right way

Rule: write the scale factor as

$$k = \frac{\text{side of the triangle you are going TO}}{\text{matching side of the triangle you are coming FROM}}$$

Then predict before you multiply:

- Going to the *smaller* triangle, $k < 1$ and every answer shrinks.
- Going to the *larger* triangle, $k > 1$ and every answer grows.
- An answer on the wrong side of the side you started from is a reversed factor — and the two wrong answers are reciprocals of each other, never near misses.

Example: $\triangle RST \sim \triangle XYZ$ with $RS = 18$, $XY = 6$ and $ST = 15$. Find YZ.
Step 1. The known pair $RS = 18$ and $XY = 6$ says XYZ is the smaller triangle, so the factor must be less than 1 and YZ must be shorter than 15.
Step 2. $k = \dfrac{6}{18} = \dfrac{1}{3}$, so $YZ = 15 \cdot \dfrac{6}{18}$. $YZ = \mathbf{5}$, which is indeed less than 15.

Try it: $\triangle RST \sim \triangle XYZ$ with $RS = 21$, $XY = 6$ and $ST = 14$. Find YZ.

check: **4**

Skill 3 — Scaling an area, not a side

Rule: if corresponding sides are in the ratio k, then

$$\frac{\text{larger area}}{\text{smaller area}} = k^2$$

An area is a product of two lengths, and *both* lengths get multiplied by k, so the area is multiplied by $k \cdot k$. Using k alone on an area is the most common scale-factor error after reversing the direction.

Example: Corresponding sides are in the ratio $3 : 7$ and the smaller triangle's area is 18. Find the larger area.
Step 1. The question moves between *areas*, so the side ratio has to be squared before it is used.
Step 2. $k = \dfrac{7}{3}$, so $k^2 = \dfrac{49}{9}$, and the larger area is $18 \cdot \dfrac{49}{9}$. The larger area is **98**.

Try it: Corresponding sides are in the ratio $2 : 5$ and the smaller triangle's area is 12. Find the larger area.

check: **75**

(!) Watch out: A reversed correspondence and a reversed scale factor both produce a clean-looking number, so re-doing the arithmetic never catches them. Catch them by size instead: decide which triangle is larger *first*, and reject any answer that moves the wrong way.